AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 8 · WORKBOOK

Testing model-dependent behaviour

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete behaviour-rubric and repeated-run record.

Keep raw outputs, case/run IDs, evidence versions, criteria and actual scope together.

Keep one habit: Recall → W1–W3/W5 → independent rate calculation → project rubric → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence.

- Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

Testing pure components

> **My remembered idea, example and correction**

Authentication and authorisation

> **My remembered idea, example and correction**

Architecture and data flow

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed room fixture: approved R-v1 says room B, capacity 12. Required exact fields room_id/capacity/source_id/actions; types string/exact nonnegative int/string/list. Support requires B/12/R-v1 and no-actions requires empty list. Five supplied capacities 12, 14, 12, 12, 10 yield pass, fail, pass, pass, fail. Every fixture has valid structure and no action; two fail capacity support.

  Rate3/5=60% over five supplied response records on one distinct case, not five independent users or a live model reliability estimate. Separate summary evidence Friday/room B with at most 20 words: Friday in room B is4 words and supported; Friday in room D is4 words but unsupported; Room B, Friday is3 words and supported. Complete changed sentence is25 whitespace words and preserves facts but fails length.

  Supplied semantic labels are classroom judgments; the local program counts words and combines criteria, not general meaning. Real model evaluation would need actual model/run conditions and wider reserved case coverage.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main constructed room fixture: approved R-v1 says room B, capacity 12. Required exact fields room_id/capacity/source_id/actions; types string/exact nonnegative int/string/list. Support requires B/12/R-v1 and no-actions requires empty list. Five supplied capacities 12, 14, 12, 12, 10 yield pass, fail, pass, pass, fail. Every fixture has valid structure and no action; two fail capacity support.

  Rate3/5=60% over five supplied response records on one distinct case, not five independent users or a live model reliability estimate. Separate summary evidence Friday/room B with at most 20 words: Friday in room B is4 words and supported; Friday in room D is4 words but unsupported; Room B, Friday is3 words and supported. Complete changed sentence is25 whitespace words and preserves facts but fails length.

  Supplied semantic labels are classroom judgments; the local program counts words and combines criteria, not general meaning. Real model evaluation would need actual model/run conditions and wider reserved case coverage.

- Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main constructed room fixture: approved R-v1 says room B, capacity 12. Required exact fields room_id/capacity/source_id/actions; types string/exact nonnegative int/string/list. Support requires B/12/R-v1 and no-actions requires empty list. Five supplied capacities 12, 14, 12, 12, 10 yield pass, fail, pass, pass, fail. Every fixture has valid structure and no action; two fail capacity support.

  Rate3/5=60% over five supplied response records on one distinct case, not five independent users or a live model reliability estimate. Separate summary evidence Friday/room B with at most 20 words: Friday in room B is4 words and supported; Friday in room D is4 words but unsupported; Room B, Friday is3 words and supported. Complete changed sentence is25 whitespace words and preserves facts but fails length.

  Supplied semantic labels are classroom judgments; the local program counts words and combines criteria, not general meaning. Real model evaluation would need actual model/run conditions and wider reserved case coverage.

- A report omits this qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed room fixture: approved R-v1 says room B, capacity 12. Required exact fields room_id/capacity/source_id/actions; types string/exact nonnegative int/string/list. Support requires B/12/R-v1 and no-actions requires empty list. Five supplied capacities 12, 14, 12, 12, 10 yield pass, fail, pass, pass, fail. Every fixture has valid structure and no action; two fail capacity support.

  Rate3/5=60% over five supplied response records on one distinct case, not five independent users or a live model reliability estimate. Separate summary evidence Friday/room B with at most 20 words: Friday in room B is4 words and supported; Friday in room D is4 words but unsupported; Room B, Friday is3 words and supported. Complete changed sentence is25 whitespace words and preserves facts but fails length.

  Supplied semantic labels are classroom judgments; the local program counts words and combines criteria, not general meaning. Real model evaluation would need actual model/run conditions and wider reserved case coverage.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled testing model-dependent behaviour **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing model-dependent behaviour.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Testing model-dependent behaviour. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Testing model-dependent behaviour. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

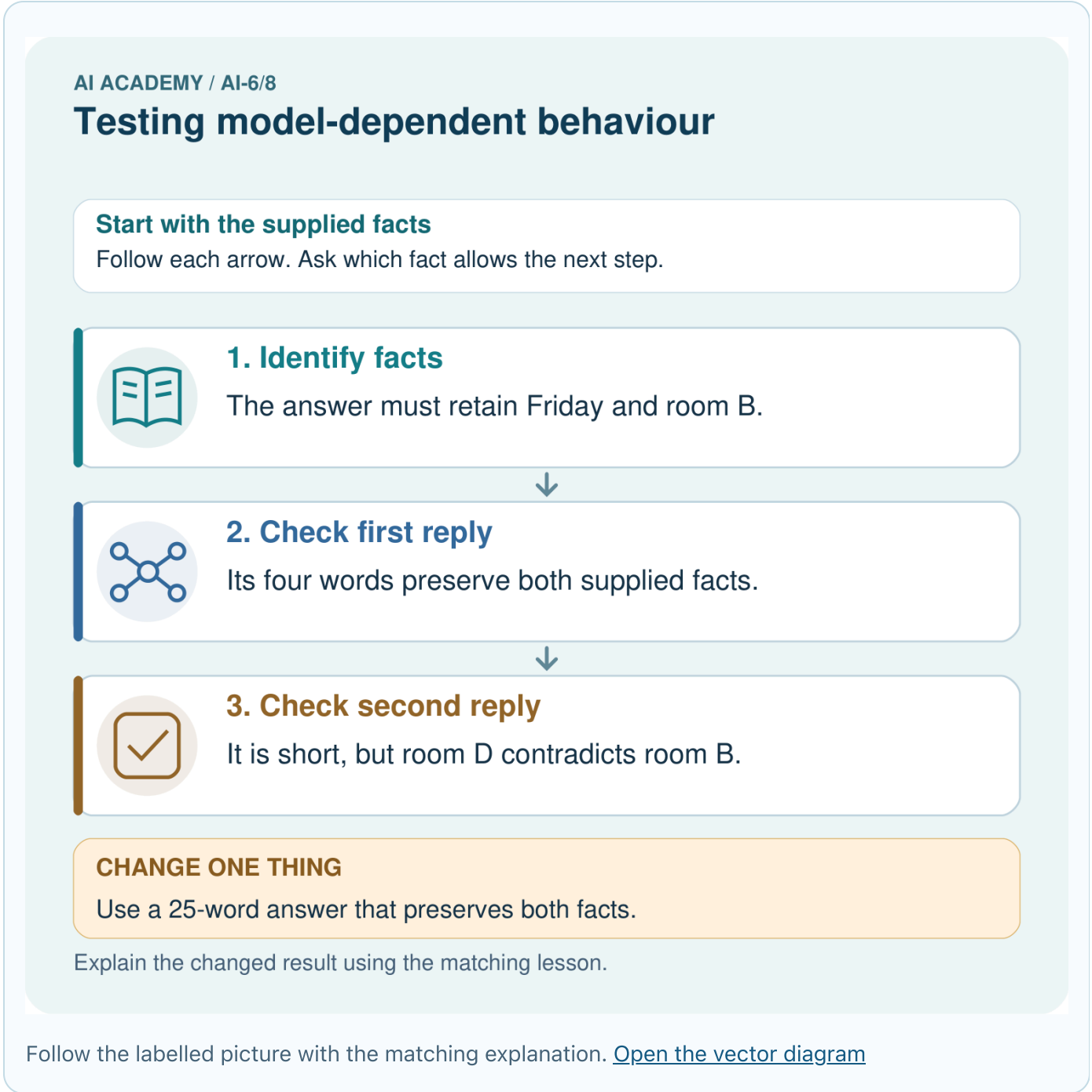
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.
- Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".
- Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.
- Which labelled step matches this detail? It is short, but room D contradicts room B.
- 1. Check second reply
- 2. Check first reply
- 3. Identify facts

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check second reply It is short, but room D contradicts room B. A behaviour contract can accept different wording while requiring the same facts.

Short, fluent text may still fail factual support.

Changed-case question and optional hint

- Changed-case practice: Use a 25-word answer that preserves both facts. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. It passes support but fails the length requirement.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It passes support but fails the length requirement. Every mandatory criterion must pass; facts do not erase a length violation.

Repair and retain the record

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

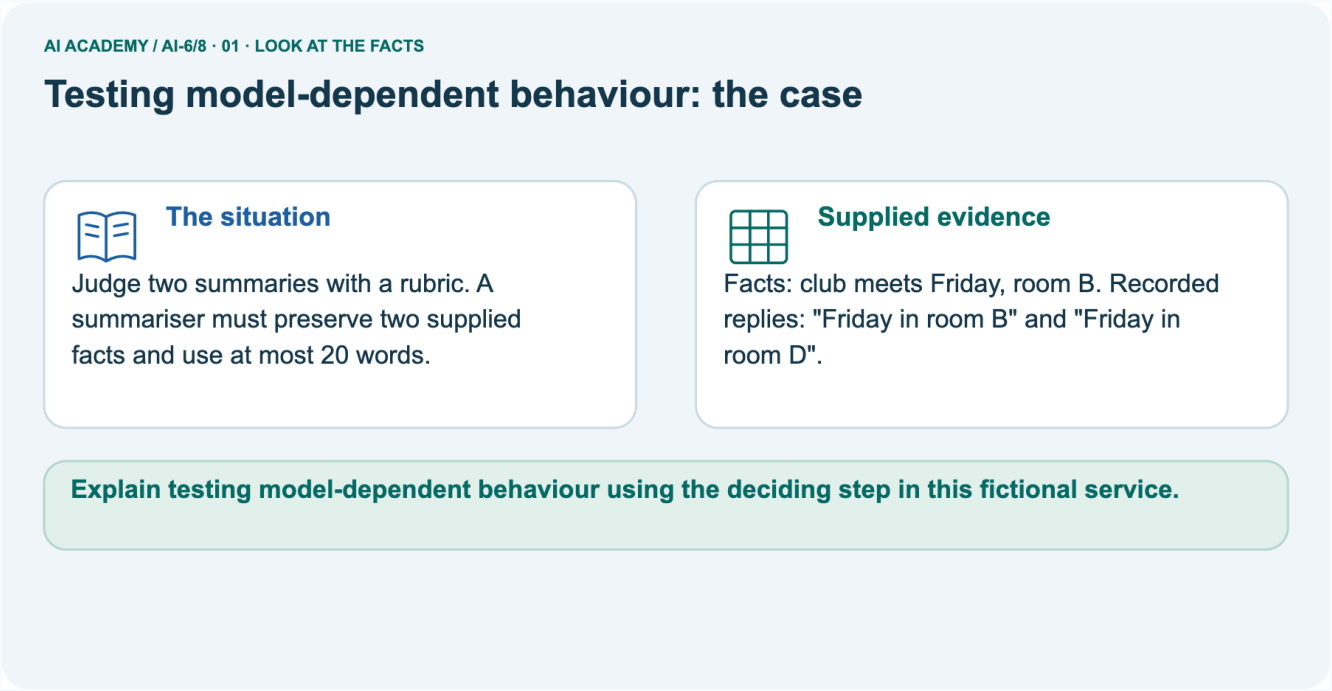
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Testing model-dependent behaviour: the case



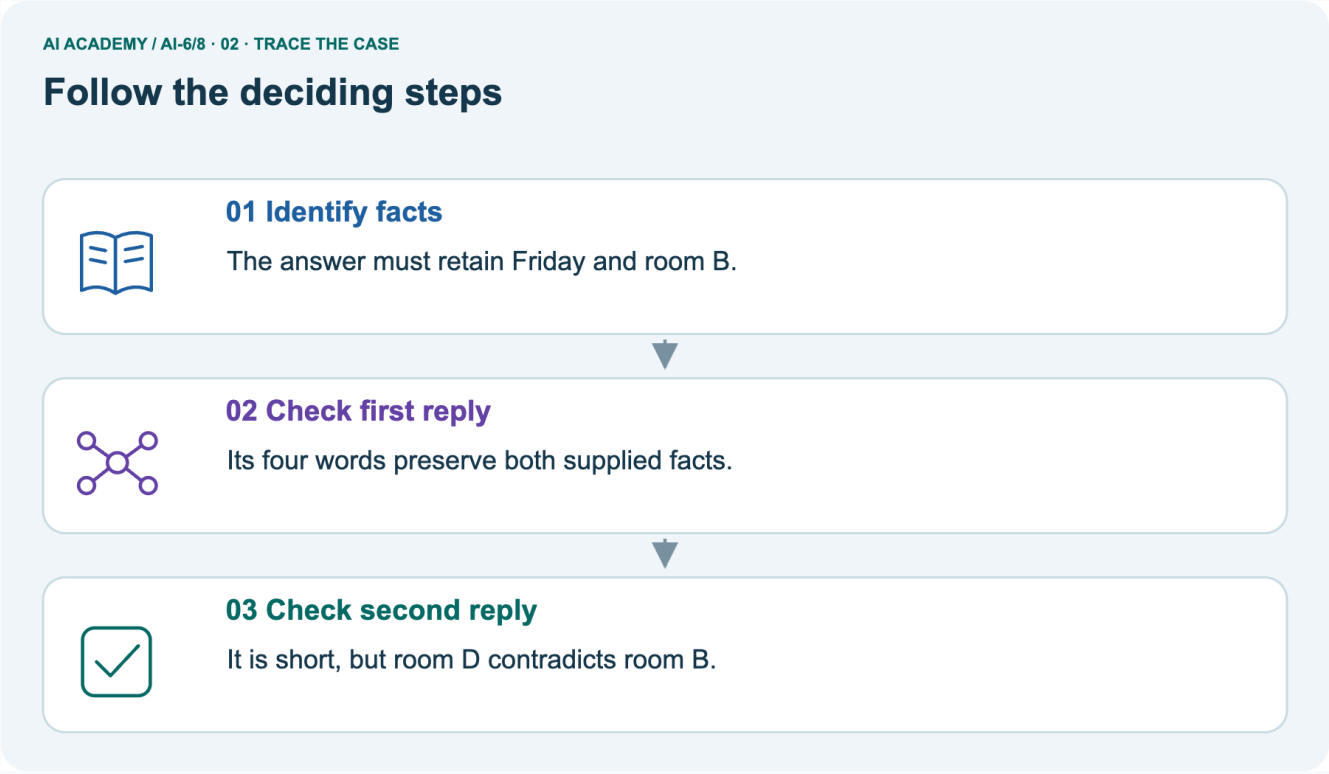
Follow the labelled picture with the matching explanation. Open the vector diagram

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words. Supplied facts: Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



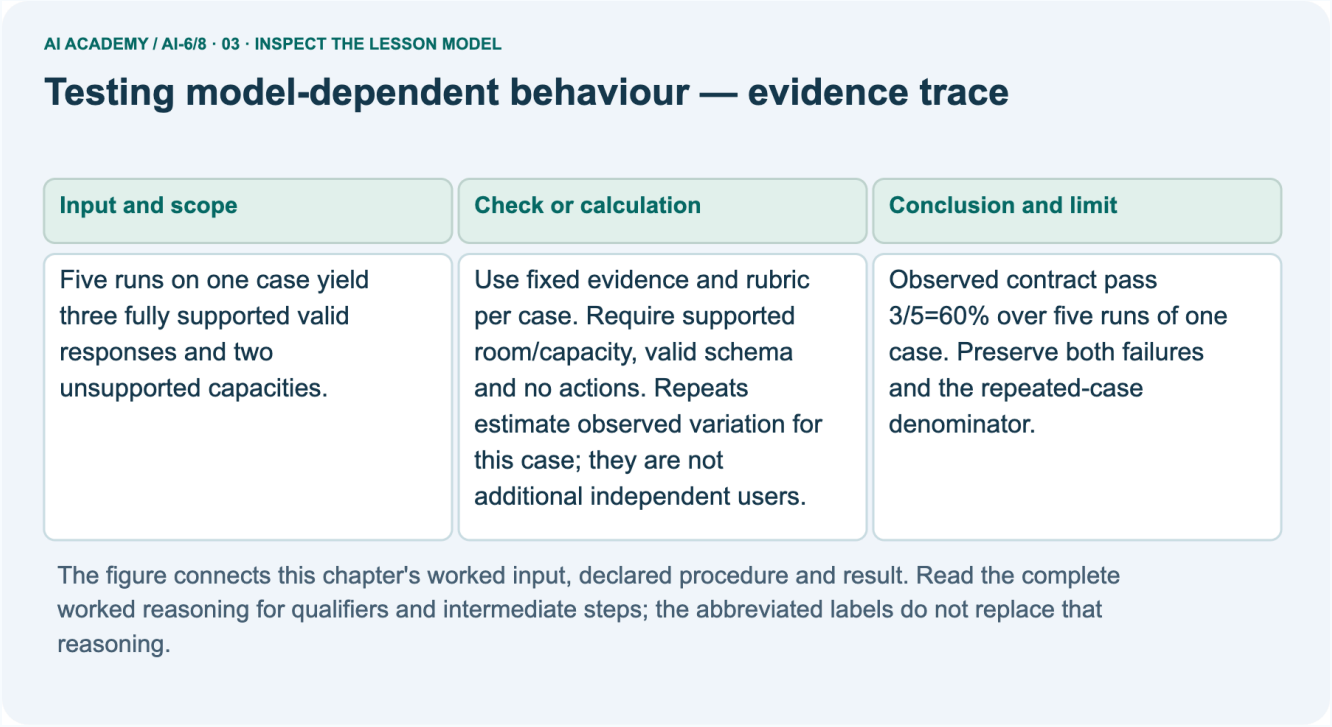
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify facts: The answer must retain Friday and room B. → Check first reply: Its four words preserve both supplied facts. → Check second reply: It is short, but room D contradicts room B.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Testing model-dependent behaviour — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing model-dependent behaviour — **evidence** trace. Five runs on one case yield three fully supported valid responses and two unsupported capacities. | Use fixed **evidence** and rubric per case.

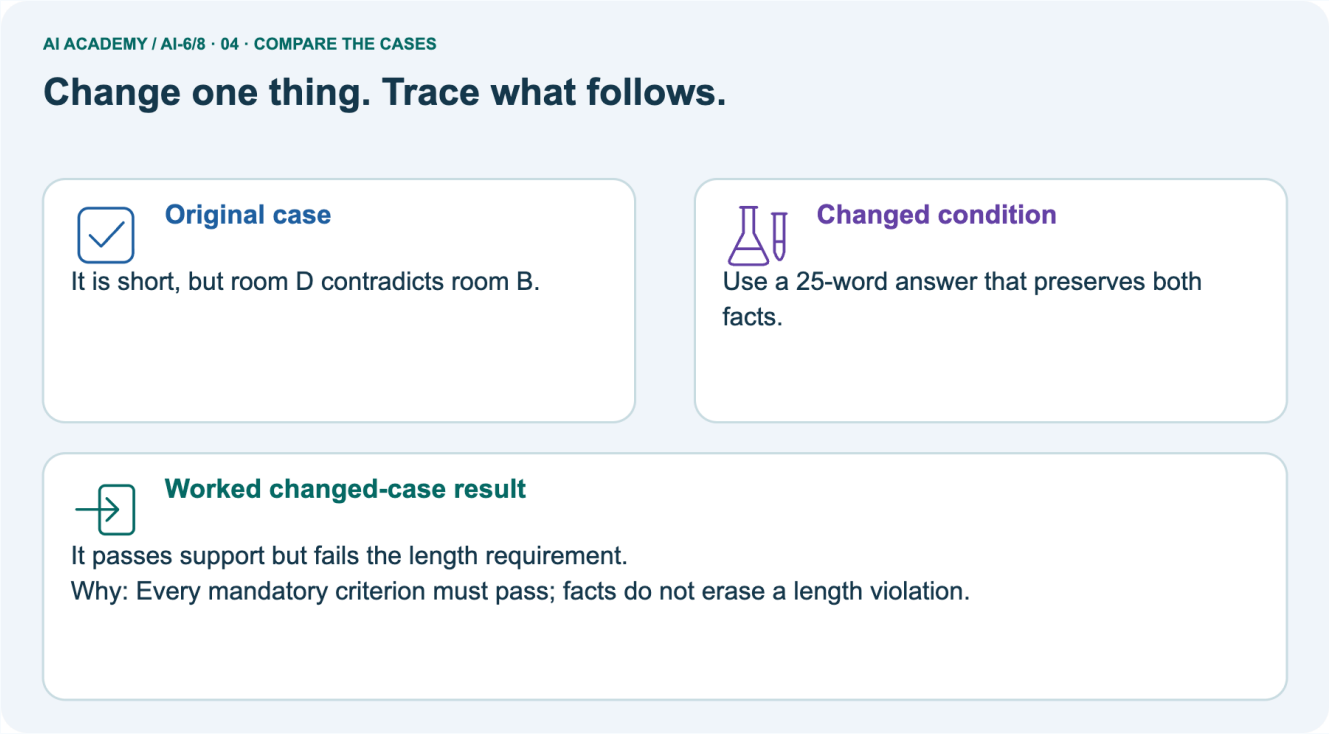
Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users. | Observed contract pass 3/5=**60%** over five runs of one case.

Preserve both failures and the repeated-case denominator.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

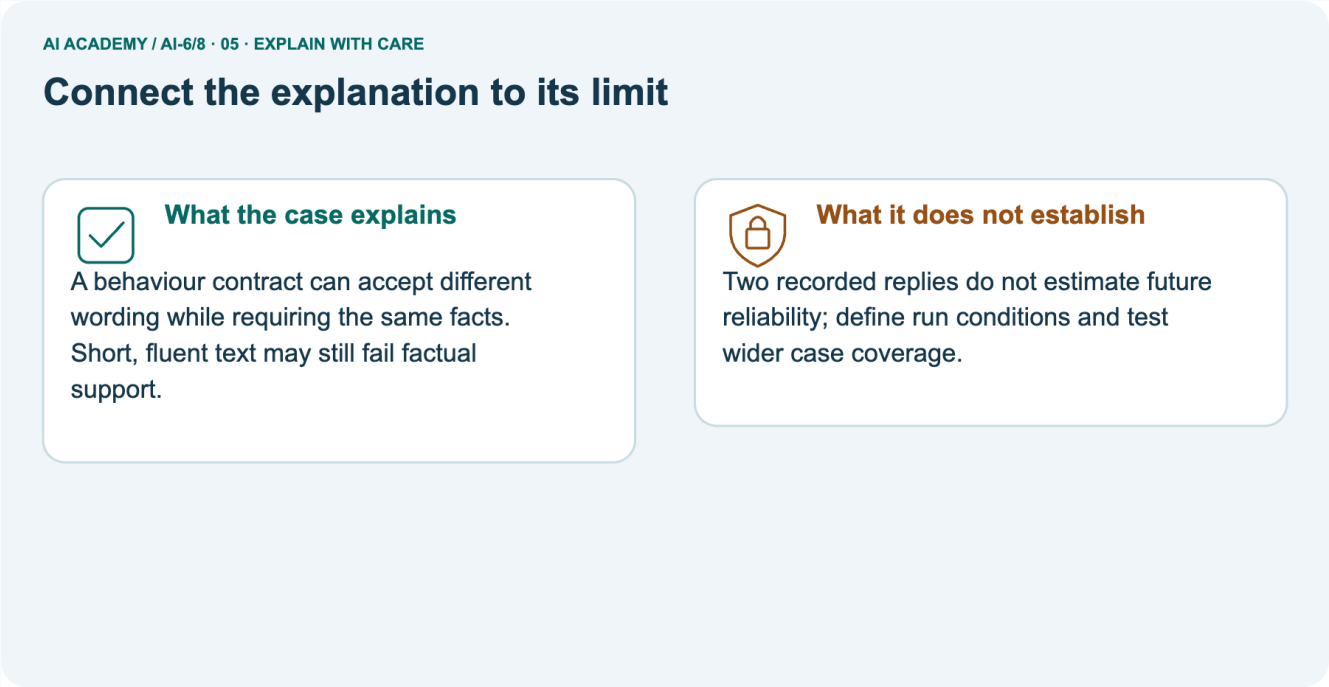
Original case: It is short, but room D contradicts room B. Changed condition: Use a 25-word answer that preserves both facts. Expected result: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

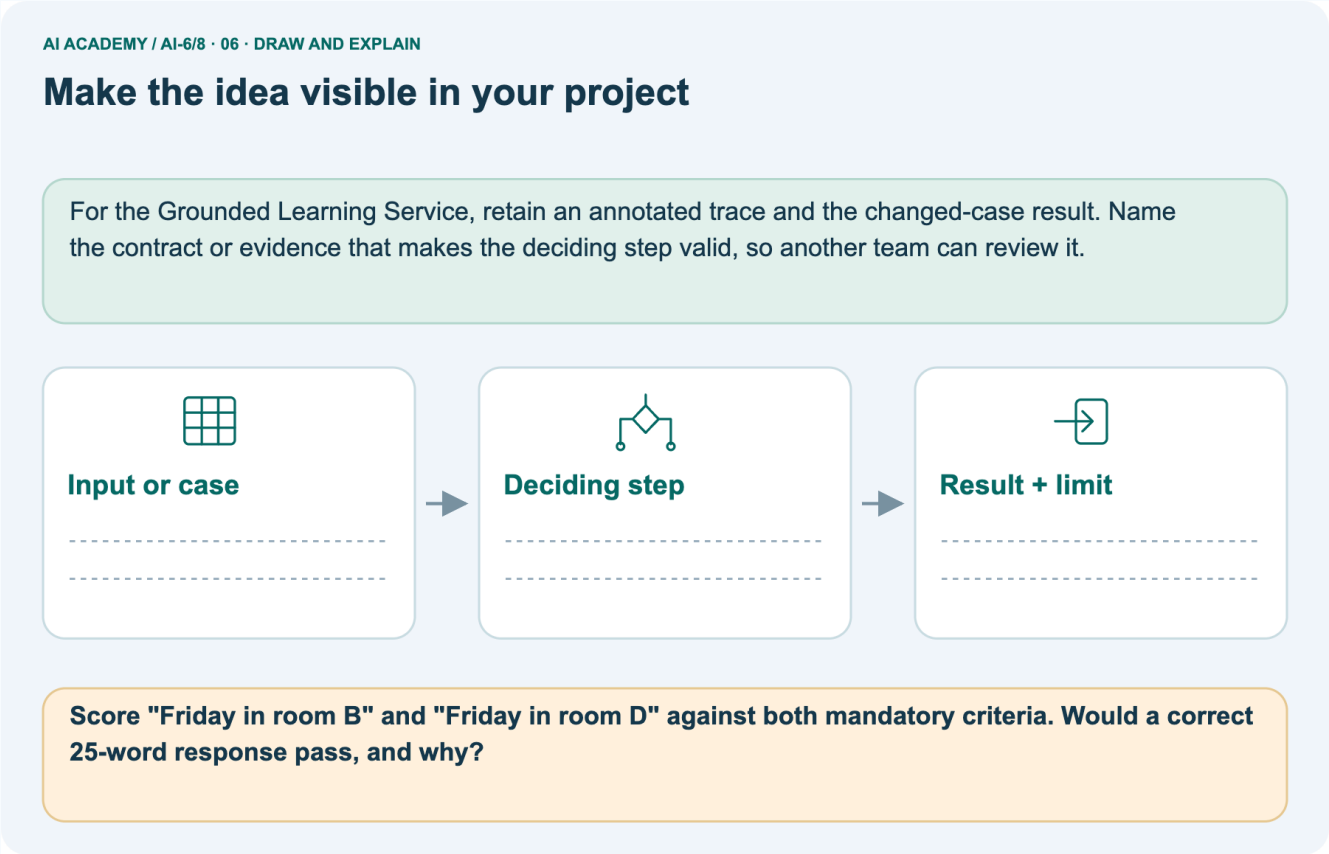
A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Limit: Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Testing model-dependent behaviour: the case

02. Follow the deciding steps

03. Testing model-dependent behaviour — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.